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ENC 3241

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Project #3: Proposal Draft

Project Title and Faculty Mentor

Project Title: Plastic invading Mosquitos: Quantifying the Atmospheric and Waterborne Microplastic arriving to Mosquito Lagoon

Faculty Mentor: Dr. Linda J Walters is a professor of Biology at the University of Central Florida. Through my research on the effects of microplastic pollution in Florida I found her name echoed in various papers and figured out that Dr. Walters directs the Coastal and Estuarine Ecology Lab and has been studying microplastic pollution in the Indian River Lagoon. That makes her the ideal mentor for the proposed project.

Project Objective or Aim

This project has the objective of quantifying the Atmospheric and Waterborne deposition of microplastics into Mosquito Lagoon and how does this vary between protected shorelines (No/low human activity) against developed shorelines (a controlled variable)? To fulfill this objective I will deploy both atmospheric collectors and grab samples at six sites, then with those I will use various quantifying techniques (as those used by Walters) to have some numbers I can then use to compare to the study done on the Indian River Lagoon (right next door). The addition of air collectors is important to control a factor that could mess up with your reasoning.

Project Background and Significance

Microplastics are now documented contaminants in every major Florida estuary. Walters (2022) reported a mean surface-water concentration of 1.47 microplastics per liter across the Indian River Lagoon. On the gulf side of Florida McEachern (2019) documented 0.25-7.0 particles per liter in Tampa Bay Surface waters and estimated that the bay holds around 4 billion microplastic particles at any given moment. Walters (2018) also conducted a study on Oysters and Atlantic mud crabs in Mosquito Lagoon accumulating Microplastics in their tissue. Janosik (2021) extended the finding to predator marine snails, suggesting that microplastics are transferred upward through Estuarine food webs (Reaching food we eat)

Until recently the assumption was that runoff discharge accounts for most of the plastic load in these systems, but Walters (2025) adds a new pivot point. Using atmospheric collectors across the Indian River Lagoon the authors measured an average fallout rate of 1224 microplastics per square meter per day which puts the atmospheric input at 1.1 trillion particles every day. Comparable to their earlier estimate of 1.4 trillion particles present in the lagoon

This proposal is grounded in the pollutant pathway framework used in contaminant studies. Which treats the Estuary as a system with measurable input, internal circulation and sinks. As mentioned previously (in the Project Objective) the idea of sampling air and water at the same sites is inspired by that, this project can use both measurements to estimate the relative magnitude on the microplastic levels in Mosquito Lagoon specifically. The significance could fill in a gap of lack of atmospheric data in Mosquito Lagoon.

The findings of this study would be useful for management. If atmospheric input in Mosquito Lagoon turns out to be similar to what Walters (2025) found in the Indian River Lagoon, that changes how mitigation should be approached. For years the focus has been on stormwater runoff but if half the microplastic load is falling from the sky, cleaning outfalls alone

is not going to fix the problem. The data could help Canaveral National Seashore and Florida environmental agencies decide where to target their resources. It also ties back to Walters (2018) which already showed oysters in Mosquito Lagoon are picking up these particles, meaning there is a seafood angle too.

Research Methods

I will sample six sites along mosquito lagoon arranged in pairs, two adjacent to the developed area (Map below for visualization) which is around M1, two inside the Canaveral National Seashore, and two at the transition between the. Each site hosts one passive atmospheric collector and one water collector. The pair ensures that most variables are kept the same while getting samples from both the air and the sea. Which is unique to this study.

Over the Fall 2026 semester I will collect samples monthly for 3-5 months, depending on the data I've gathered at the start. At each visit both collectors are dumped into glass containers and then redeployed, to ensure we have a vial ready to analyze. The glass bottles have yet to be determined but it would have to match the expected sourcing from the collectors. All samples are transported to the lab, doing our best to keep it uncontaminated, and ensures that our cadence matches that used by Walters (2022)

Once the vials are in our possession we will conduct laboratory analysis to determine the pollution. While field analysis has changed over the years the laboratory side of things is properly transcribed in the studies Walters (2022). The laboratory will then yield results of the particles. We will count them and classify them depending on what type of microplastic it is.

Once we've extracted the numbers we will analyze them depending on their means. Atmospheric deposition will be in terms of microplastics per square meter per day, and water collectors in microplastics per liter. We will then compare these metrics across the different sites

and month, using this we will determine the relationship between Atmospheric and water microplastic pollution and how they work with each other.

The timeline starts in the Fall 2026 semester. Where the first 2 weeks will be used to make sure we have everything we need (permission, lab, equipment, proper etiquette), and then it will be followed by a week of seeing and deploying the collectors. Then it's followed by 3-5 months of processing the data and analysing it. The times vary since depending on the results we could conclude something earlier.

Expected Outcome

This project will produce various deliverables. Primarily our findings of the measurements at six Mosquito Lagoon sites across the 3-5 months which will be presented in the form of a dataset. The first of its kind. This great research paper could be extremely useful to said authorities and help push the environmental task force to find correlation between different events, current or past, that have happened. Then we will also produce a research paper fit for submission at a peer-reviewed journal such as environmental or ecological ones, where we could have Dr. Walters as a co-author. Finally we will yield a brief summary of our findings to provide an easy source of information for anyone trying to follow through. (In the form of an infographic containing our extraction points and different levels of microplastics.)

The results will be shared in various different ways. Primarily to garner some reputable feedback it would be shared in the journal as mentioned above, and it would hopefully lead to further work done. I would also use any available UCF tools and social media to spread awareness if our findings require it. The nature of the sharing will be ecological more than anything else.

The knowledge gained from this study will be niche but fill in a gap in the current studies of pollution in Florida's estuaries. The research at Mosquito Lagoon will be extremely beneficial to the studies done at UCF by others by adding onto current research the atmospheric and waterborne microplastics at a body of water near where another study happened. It would also fundamentally help everyone to understand the current levels of pollution, its daily and long term effects, and where it could progress with current analysed data.

Beyond the academic side, the dataset itself could be handed off to UCF's Coastal and Estuarine Ecology Lab so future students can build on it without starting from scratch.

Literature Review

Busch, Sidney J., et al. "Contribution of Stormwater Outfalls to Microplastic Pollution in a Subtropical Estuary Using Data Collected with the Assistance of Citizen Scientists." Environments, vol. 10, no. 10, 2023, article 181, https://www.mdpi.com/2076-3298/10/10/181.

Kleinschmidt, Julia M., and Alexis M. Janosik. "Microplastics in Florida, United States: A Case Study of Quantification and Characterization with Intertidal Snails." Frontiers in Ecology and Evolution, vol. 9, 2021, article 645727, https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2021.645727/full.

McEachern, Kinsley, et al. "Microplastics in Tampa Bay, Florida: Abundance and Variability in Estuarine Waters and Sediments." Marine Pollution Bulletin, vol. 148, 2019, pp. 97–106, https://pubmed.ncbi.nlm.nih.gov/31422308/.

Waite, Heidi R., Melinda J. Donnelly, and Linda J. Walters. "Quantity and Types of Microplastics in the Organic Tissues of the Eastern Oyster Crassostrea virginica and

Atlantic Mud Crab Panopeus herbstii from a Florida Estuary." Marine Pollution Bulletin, vol. 129, no. 1, 2018, pp. 179–185, https://pubmed.ncbi.nlm.nih.gov/29680536/.

Walters, Linda J., et al. "Quantifying Spatial and Temporal Trends of Microplastic Pollution in Surface Water and in the Eastern Oyster Crassostrea virginica for a Dynamic Florida Estuary." Environments, vol. 9, no. 10, 2022, article 131, https://www.mdpi.com/2076-3298/9/10/131.

Walters, Linda J., et al. "Raining Plastics: Quantification of Atmospheric Deposition of Plastic and Anthropogenic Particles into an Estuary of National Significance with the Assistance of Citizen Scientists." Environments, vol. 12, no. 11, 2025, article 424, https://www.mdpi.com/2076-3298/12/11/424.

Preliminary Work and Experience

I don't have any preliminary work and experience in this field. But I do have external knowledge that could be beneficial. I have been studying microplastic effects on humans for a while (from where they come from, how they affect us, where they recide….) and that interest led me to this study while brainstorming for ideas (Outside of the fact that UCF has partaken in this research). I have also written research papers before, which although not direct biology implications it shows that I have what is needed to follow the correct framework to at least derive some findings.

The research papers I have written before were not in biology but they required the same skills, finding sources, building an argument, and putting together a literature review. I have also been following microplastic research outside of class for over a year, which is what got me into this topic in the first place.

8. IRB/ IACUC Statement

This project does not require any of both approvals as it involves no human or animal subject. It's purely ecological and there will be no direct interaction with protected animals. Sampling will consist of water and air.

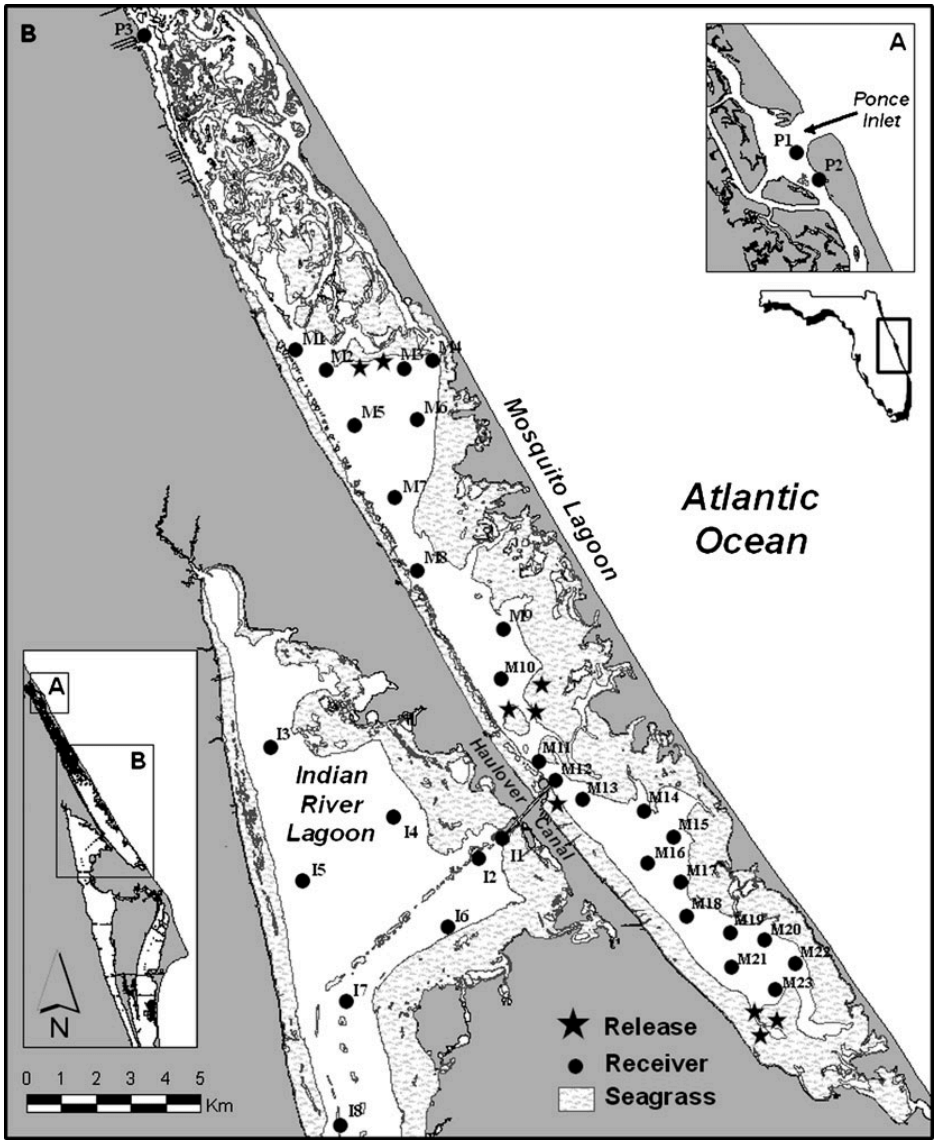
9. Budget

I am requesting a total of $800.00. The funds will be used as follows:

Item	**Quantity**	**Cost**	**Notes**
Atmospheric collectors	6	$200	Passive collectors deployed at each of the six sites
Sample containers (glass)	~24	$200	Monthly redeployment across 3-5 months of sampling
Travel	—	$200	Gas and vehicle depreciation for site visits to Mosquito Lagoon (unless transportation is provided)
Documentation & miscellaneous	—	$200	Paperwork, lab supplies, and a small reimbursement buffer
Total		**$800**	

10. Figures and Tables

Figure 1 (below) shows the proposed sampling locations within Mosquito Lagoon. Six sites are arranged in three pairs: two near the developed Titusville shoreline, two in the transition zone, and two within the protected boundaries of Canaveral National Seashore.


B
A
Ponce Inlet
P1
P2
P3
Mosquito Lagoon
Atlantic Ocean
Indian River Lagoon
Haulover Canal
M1
M2
M3
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M16
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M20
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I1
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Release
Receiver
Seagrass
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